

Sequences & Series

ANSWER KEY



Arithmetic sequences: finding terms

- 1 Find the 12th term of the arithmetic sequence with $a = 5$, $d = 3$.

$$a_{12} = 5 + 11(3) = 5 + 33 = 38.$$

- 2 Find the 20th term of the arithmetic sequence with $a = -4$, $d = 2$.

$$a_{20} = -4 + 19(2) = -4 + 38 = 34.$$

- 3 Find the 15th term of the arithmetic sequence with $a = 100$, $d = -6$.

$$a_{15} = 100 + 14(-6) = 100 - 84 = 16.$$

- 4 Write the first four terms of the arithmetic sequence with $a = 7$, $d = -2$.

$$7, 5, 3, 1.$$

Arithmetic sequences: finding a and d

- 5 The 4th term of an arithmetic sequence is 17 and the 9th term is 42. Find the first term a and common difference d .

$$a + 3d = 17 \text{ and } a + 8d = 42. \text{ Subtracting: } 5d = 25 \Rightarrow d = 5. \text{ Then } a = 17 - 15 = 2.$$

- 6 An arithmetic sequence has $a_1 = 10$ and $a_6 = 30$. Find the common difference d .

$$a_6 = a_1 + 5d = 10 + 5d = 30 \Rightarrow 5d = 20 \Rightarrow d = 4.$$

- 7 An arithmetic sequence has $a_3 = 11$ and $a_7 = 27$. Find the common difference d .

$$a_7 - a_3 = 4d = 16 \Rightarrow d = 4.$$

Arithmetic series: the sum formula

- 8 Find the sum of the first 20 terms of the arithmetic sequence with $a = 3$, $d = 4$, using

$$S_n = \frac{n}{2}(2a + (n-1)d).$$
$$S_{20} = \frac{20}{2}(6 + 19(4)) = 10(6 + 76) = 10(82) = 820.$$

- 9 Find the sum of the first 15 terms of the arithmetic sequence with $a = 50$, $d = -3$.

$$S_{15} = \frac{15}{2}(100 + 14(-3)) = 7.5(100 - 42) = 7.5(58) = 435.$$

- 10 Find the sum of the arithmetic series $2 + 5 + 8 + \dots + 50$.

Here $a = 2$, $d = 3$; $50 = 2 + (n - 1)3 \Rightarrow n = 17$. $S_{17} = \frac{17}{2}(2 + 50) = 17(26) = 442$.

- 11 Find the sum of the first 10 positive multiples of 6.

These form an arithmetic sequence with $a = 6$, $d = 6$, $n = 10$.
 $S_{10} = \frac{10}{2}(12 + 9(6)) = 5(12 + 54) = 5(66) = 330$.

Geometric sequences: finding terms

- 12 Find the 6th term of the geometric sequence with $a = 3$, $r = 2$.

$$a_6 = 3(2)^5 = 3(32) = 96.$$

- 13 Find the 5th term of the geometric sequence with $a = 100$, $r = 0.5$.

$$a_5 = 100(0.5)^4 = 100(0.0625) = 6.25.$$

- 14 Find the 4th term of the geometric sequence with $a = 2$, $r = -3$.

$$a_4 = 2(-3)^3 = 2(-27) = -54.$$

- 15 Write the first four terms of the geometric sequence with $a = 5$, $r = 3$.

$$5, 15, 45, 135.$$

Geometric sequences: finding a and r

- 16 The 2nd term of a geometric sequence is 6 and the 5th term is 162. Find r and a .

$$ar = 6 \text{ and } ar^4 = 162. \text{ Dividing: } r^3 = 27 \Rightarrow r = 3. \text{ Then } a = 6/3 = 2.$$

- 17 A geometric sequence has $a_1 = 4$ and $a_4 = 108$. Find r .

$$a_4 = a_1 r^3 = 4r^3 = 108 \Rightarrow r^3 = 27 \Rightarrow r = 3.$$

- 18 A geometric sequence has 3rd term 12 and 6th term 96. Find a and r .

$$ar^2 = 12 \text{ and } ar^5 = 96. \text{ Dividing: } r^3 = 8 \Rightarrow r = 2. \text{ Then } a = 12/4 = 3.$$

Geometric series: the finite sum

- 19 Find the sum of the first 6 terms of the geometric series with $a = 3$, $r = 2$, using $S_n = \frac{a(r^n - 1)}{r - 1}$.

$$S_6 = \frac{3(2^6 - 1)}{2 - 1} = 3(63) = 189.$$

- 20 Find the sum of the first 5 terms of the geometric series with $a = 100$, $r = 0.5$.

$$S_5 = \frac{100(1 - 0.5^5)}{1 - 0.5} = \frac{100(0.96875)}{0.5} = 193.75.$$

- 21 Find the sum of the first 4 terms of the geometric series with $a = 2$, $r = -3$.

$$S_4 = \frac{2(1 - (-3)^4)}{1 - (-3)} = \frac{2(1 - 81)}{4} = \frac{-160}{4} = -40.$$

Infinite geometric series

- 22 Find the sum of the infinite geometric series with $a = 4$, $r = \frac{1}{2}$.

$$S = \frac{a}{1 - r} = \frac{4}{1 - \frac{1}{2}} = 8.$$

- 23 Find the sum of the infinite geometric series with $a = 9$, $r = \frac{1}{3}$.

$$S = \frac{9}{1 - \frac{1}{3}} = \frac{9}{\frac{2}{3}} = 13.5.$$

- 24 Find the sum of the infinite geometric series $8 - 4 + 2 - 1 + \dots$.

$$\text{Here } a = 8, r = -\frac{1}{2}. S = \frac{8}{1 - (-\frac{1}{2})} = \frac{8}{1.5} = \frac{16}{3}.$$

- 25 Express the repeating decimal $0.333\dots$ as a fraction using an infinite geometric series with $a = \frac{3}{10}$, $r = \frac{1}{10}$.

$$S = \frac{3/10}{1 - 1/10} = \frac{3/10}{9/10} = \frac{3}{9} = \frac{1}{3}.$$

Sigma notation

- 26 Evaluate $\sum_{k=1}^5 (2k + 1)$.

Terms: 3, 5, 7, 9, 11. Sum = 35.

- 27 Evaluate $\sum_{k=1}^4 3^k$.

$$3 + 9 + 27 + 81 = 120.$$

- 28 Use $\sum_{k=1}^n k = \frac{n(n+1)}{2}$ to evaluate $\sum_{k=1}^{10} k$.

$$\frac{10(11)}{2} = 55.$$

Recursive sequences

- 29 A sequence satisfies $a_1 = 2$ and $a_{n+1} = a_n + 5$. Find a_5 .
 $a_1 = 2, a_2 = 7, a_3 = 12, a_4 = 17, a_5 = 22$.
- 30 A sequence satisfies $a_1 = 3$ and $a_{n+1} = 2a_n - 1$. Find a_4 .
 $a_1 = 3, a_2 = 5, a_3 = 9, a_4 = 17$.
- 31 A sequence satisfies $a_1 = 1, a_2 = 1$, and $a_n = a_{n-1} + a_{n-2}$ for $n \geq 3$. Find a_6 .
 $a_3 = 2, a_4 = 3, a_5 = 5, a_6 = 8$.
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Identifying arithmetic and geometric sequences

- 32 State whether the sequence 4, 9, 14, 19, ... is arithmetic or geometric, and give the common difference or ratio.
Arithmetic, with common difference $d = 5$.
- 33 State whether the sequence 2, 6, 18, 54, ... is arithmetic or geometric, and give the common difference or ratio.
Geometric, with common ratio $r = 3$.
- 34 State whether the sequence 100, 90, 80, 70, ... is arithmetic or geometric, and find its 10th term.
Arithmetic, $d = -10$. $a_{10} = 100 + 9(-10) = 100 - 90 = 10$.
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Reading a sequence from its terms

- 35 The bar chart shows the first five terms of a sequence.
- a Determine whether the sequence is arithmetic or geometric, and find the common ratio or difference. Geometric — each term is double the previous one, so $r = 2$.
 - b Find the 8th term. $a_8 = 2 \times 2^7 = 2 \times 128 = 256$.
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Word problems: sequences in context

- 36 This question has two parts. An employee's starting monthly salary is 4000 dollars, and it increases by a fixed amount of 150 dollars each month.
- a Write a formula for the salary a_n in month n (with $a_1 = 4000$). $a_n = 4000 + (n - 1)(150)$.
 - b Find the salary in month 12. $a_{12} = 4000 + 11(150) = 4000 + 1650 = 5650$ dollars.

- 37 This question has two parts. A bacteria culture starts with 500 cells and triples every hour. Let a_n be the number of cells after n hours, with $a_0 = 500$.
- a Write a formula for a_n . $a_n = 500 \times 3^n$.
- b Find the number of cells after 4 hours. $a_4 = 500 \times 3^4 = 500 \times 81 = 40500$ cells.
-

MCQ — Arithmetic sequences: finding terms

- 38 Find the 14th term of the arithmetic sequence with $a = 6$, $d = 4$.

- a. 62
- b. 54
- c. 50
- d. 58 ✓

$$T_{14} = 6 + 13(4) = 58.$$

- 39 Find the 18th term of the arithmetic sequence with $a = -7$, $d = 3$.

- a. 50
- b. 41
- c. 47
- d. 44 ✓

$$T_{18} = -7 + 17(3) = 44.$$

- 40 Find the 12th term of the arithmetic sequence with $a = 80$, $d = -5$.

- a. 20
- b. 30
- c. 25 ✓
- d. -25

$$T_{12} = 80 + 11(-5) = 25.$$

41 Write the first four terms of the arithmetic sequence with $a = 9$, $d = -3$.

a. 9, 12, 15, 18

b. 9, 6, 3, 0 ✓

c. 9, 3, 0, -3

d. 9, -3, -15, -27

Each term subtracts 3: 9, 6, 3, 0.

42 Find the 25th term of the arithmetic sequence with $a = 2$, $d = 7$.

a. 177

b. 170 ✓

c. 175

d. 163

$$T_{25} = 2 + 24(7) = 170.$$

MCQ — Arithmetic sequences: finding a and d

43 The 5th term of an arithmetic sequence is 19 and the 10th term is 44. Find the first term a and common difference d .

a. $a = -5$, $d = 5$

b. $a = -1$, $d = 4$

c. $a = 4$, $d = 5$

d. $a = -1$, $d = 5$ ✓

$$d = \frac{44 - 19}{5} = 5; a = 19 - 4(5) = -1.$$

44 An arithmetic sequence has $a_1 = 8$ and $a_7 = 32$. Find the common difference d .

a. $d = 4$ ✓

b. $d = 24$

c. $d = 4.8$

d. $d = 3$

$$d = \frac{32 - 8}{6} = 4.$$

45 An arithmetic sequence has $a_2 = 9$ and $a_6 = 29$. Find the common difference d .

a. $d = 5.5$

b. $d = 4$

c. $d = 20$

d. $d = 5$ ✓

$$d = \frac{29 - 9}{4} = 5.$$

46 The 3rd term of an arithmetic sequence is 15 and the 8th term is 45. Find the first term a and common difference d .

a. $a = 3, d = 5$

b. $a = 3, d = 6$ ✓

c. $a = 6, d = 3$

d. $a = 9, d = 6$

$$d = \frac{45 - 15}{5} = 6; a = 15 - 2(6) = 3.$$

MCQ — Arithmetic series: the sum formula

- 47 Find the sum of the first 25 terms of the arithmetic sequence with $a = 4$, $d = 3$, using $S_n = \frac{n}{2}(2a + (n - 1)d)$.

- a. 1,075
- b. 960
- c. 1,000 ✓
- d. 925

$$S_{25} = \frac{25}{2}(8 + 24(3)) = 12.5(80) = 1,000.$$

- 48 Find the sum of the first 18 terms of the arithmetic sequence with $a = 60$, $d = -4$.

- a. 540
- b. 456
- c. 468 ✓
- d. 480

$$S_{18} = \frac{18}{2}(120 + 17(-4)) = 9(52) = 468.$$

- 49 Find the sum of the arithmetic series $3 + 7 + 11 + \cdots + 59$.

- a. 450
- b. 434
- c. 465 ✓
- d. 480

$$n = 15 \text{ terms; } S_{15} = \frac{15}{2}(6 + 14(4)) = 7.5(62) = 465.$$

50 Find the sum of the first 12 positive multiples of 7.

a. 504

b. 546 ✓

c. 462

d. 588

$$7(1 + 2 + \cdots + 12) = 7(78) = 546.$$

51 Find the sum of the first 30 terms of the arithmetic sequence with $a = 1$, $d = 2$ (the odd numbers).

a. 960

b. 930

c. 870

d. 900 ✓

$$S_{30} = \frac{30}{2}(2 + 29(2)) = 15(60) = 900.$$

MCQ — Geometric sequences: finding terms

52 Find the 7th term of the geometric sequence with $a = 2$, $r = 3$.

a. 1,458 ✓

b. 729

c. 4,374

d. 486

$$T_7 = 2(3^6) = 1,458.$$

53 Find the 6th term of the geometric sequence with $a = 200$, $r = 0.5$.

- a. 25
- b. 3.125
- c. 12.5
- d. 6.25 ✓

$$T_6 = 200(0.5)^5 = 6.25.$$

54 Find the 5th term of the geometric sequence with $a = 3$, $r = -2$.

- a. 24
- b. 48 ✓
- c. -24
- d. -48

$$T_5 = 3(-2)^4 = 3(16) = 48.$$

55 Write the first four terms of the geometric sequence with $a = 4$, $r = 2$.

- a. 4, 6, 8, 10
- b. 4, 16, 32, 64
- c. 4, 8, 12, 16
- d. 4, 8, 16, 32 ✓

Each term doubles: 4, 8, 16, 32.

MCQ — Geometric sequences: finding a and r

56 The 2nd term of a geometric sequence is 8 and the 5th term is 64. Find r and a .

a. $r = 2, a = 8$

b. $r = 4, a = 2$

c. $r = 2, a = 4$ ✓

d. $r = 8, a = 1$

$$r^3 = 64 \div 8 = 8 \Rightarrow r = 2; a = 8 \div r = 4.$$

57 A geometric sequence has $a_1 = 5$ and $a_4 = 135$. Find r .

a. $r = 9$

b. $r = 27$

c. $r = 3$ ✓

d. $r = 2$

$$r^3 = 135 \div 5 = 27 \Rightarrow r = 3.$$

58 A geometric sequence has 3rd term 18 and 6th term 486. Find a and r .

a. $a = 18, r = 3$

b. $a = 6, r = 3$

c. $a = 2, r = 3$ ✓

d. $a = 2, r = 9$

$$r^3 = 486 \div 18 = 27 \Rightarrow r = 3; a = 18 \div r^2 = 2.$$

59 A geometric sequence has 2nd term 4 and 5th term -108 . Find r .

a. $r = 3$

b. $r = -27$

c. $r = -3$ ✓

d. $r = -9$

$$r^3 = -108 \div 4 = -27 \Rightarrow r = -3.$$

60 A geometric sequence has $a_1 = 6$ and $a_4 = -162$. Find r .

a. $r = 3$

b. $r = -9$

c. $r = -3$ ✓

d. $r = -27$

$$r^3 = -162 \div 6 = -27 \Rightarrow r = -3.$$

MCQ — Geometric series: the finite sum

61 Find the sum of the first 7 terms of the geometric series with $a = 2$, $r = 2$, using $S_n = \frac{a(r^n - 1)}{r - 1}$.

a. 256

b. 508

c. 254 ✓

d. 127

$$S_7 = \frac{2(2^7 - 1)}{2 - 1} = 2(127) = 254.$$

62 Find the sum of the first 6 terms of the geometric series with $a = 200$, $r = 0.5$.

a. 196.875

b. 400

c. 396.875

d. 393.75 ✓

$$S_6 = \frac{200(1 - 0.5^6)}{1 - 0.5} = 393.75.$$

63 Find the sum of the first 5 terms of the geometric series with $a = 3$, $r = -2$.

- a. 93
- b. 63
- c. -33
- d. 33 ✓

$$S_5 = \frac{3((-2)^5 - 1)}{-2 - 1} = \frac{3(-33)}{-3} = 33.$$

64 Find the sum of the first 8 terms of the geometric series with $a = 1$, $r = 2$.

- a. 255 ✓
- b. 128
- c. 254
- d. 256

$$S_8 = \frac{1(2^8 - 1)}{2 - 1} = 255.$$

MCQ — Infinite geometric series

65 Find the sum of the infinite geometric series with $a = 6$, $r = \frac{1}{3}$.

- a. 18
- b. 6.75
- c. 8
- d. 9 ✓

$$S = \frac{6}{1 - \frac{1}{3}} = \frac{6}{\frac{2}{3}} = 9.$$

66 Find the sum of the infinite geometric series with $a = 12$, $r = \frac{1}{4}$.

a. 14

b. 16 ✓

c. 48

d. 15

$$S = \frac{12}{1 - \frac{1}{4}} = \frac{12}{\frac{3}{4}} = 16.$$

67 Find the sum of the infinite geometric series $9 - 3 + 1 - \frac{1}{3} + \dots$.

a. 4.5

b. 9

c. 6

d. 6.75 ✓

$$a = 9, r = -\frac{1}{3}; S = \frac{9}{1 + \frac{1}{3}} = \frac{9}{\frac{4}{3}} = 6.75.$$

68 Express the repeating decimal $0.777\dots$ as a fraction using an infinite geometric series with $a = 0.7$, $r = 0.1$.

a. $\frac{7}{90}$

b. $\frac{77}{100}$

c. $\frac{7}{9}$ ✓

d. $\frac{7}{10}$

$$S = \frac{0.7}{1 - 0.1} = \frac{0.7}{0.9} = \frac{7}{9}.$$

69 Find the sum of the infinite geometric series with $a = 10$, $r = \frac{2}{5}$.

a. $\frac{50}{3}$ ✓

b. $\frac{25}{3}$

c. 14

d. 25

$$S = \frac{10}{1 - \frac{2}{5}} = \frac{10}{3/5} = \frac{50}{3}.$$

MCQ — Sigma notation

70 Evaluate $\sum_{k=1}^6 (3k - 1)$.

a. 51

b. 54

c. 57 ✓

d. 63

$$2 + 5 + 8 + 11 + 14 + 17 = 57.$$

71 Evaluate $\sum_{k=1}^5 2^k$.

a. 60

b. 62 ✓

c. 63

d. 32

$$2 + 4 + 8 + 16 + 32 = 62.$$

72 Use $\sum_{k=1}^n k = \frac{n(n+1)}{2}$ to evaluate $\sum_{k=1}^{15} k$.

a. 120 ✓

b. 110

c. 105

d. 136

$$\frac{15(16)}{2} = 120.$$

73 Evaluate $\sum_{k=2}^5 k^2$.

a. 55

b. 30

c. 50

d. 54 ✓

$$4 + 9 + 16 + 25 = 54.$$

MCQ — Recursive sequences

74 A sequence satisfies $a_1 = 3$ and $a_{n+1} = a_n + 7$. Find a_5 .

a. 31 ✓

b. 38

c. 28

d. 24

$$3, 10, 17, 24, 31: a_5 = 31.$$

75 A sequence satisfies $a_1 = 4$ and $a_{n+1} = 3a_n - 2$. Find a_4 .

- a. 10
- b. 244
- c. 28
- d. 82 ✓

4, 10, 28, 82: $a_4 = 82$.

76 A sequence satisfies $a_1 = 1$, $a_2 = 2$, and $a_n = a_{n-1} + a_{n-2}$ for $n \geq 3$. Find a_7 .

- a. 21 ✓
- b. 34
- c. 8
- d. 13

1, 2, 3, 5, 8, 13, 21: $a_7 = 21$.

77 A sequence satisfies $a_1 = 2$ and $a_{n+1} = 2a_n + 1$. Find a_4 .

- a. 11
- b. 23 ✓
- c. 5
- d. 47

2, 5, 11, 23: $a_4 = 23$.

MCQ — Identifying arithmetic and geometric sequences

78 State whether 6, 11, 16, 21, ... is arithmetic or geometric, and give the common difference or ratio.

a. Arithmetic, $d = 5$ ✓

b. Geometric, $r = \frac{11}{6}$

c. Geometric, $r = 5$

d. Arithmetic, $d = 6$

Each term adds 5: arithmetic, $d = 5$.

79 State whether 3, 9, 27, 81, ... is arithmetic or geometric, and give the common difference or ratio.

a. Geometric, $r = 6$

b. Arithmetic, $d = 6$

c. Arithmetic, $d = 3$

d. Geometric, $r = 3$ ✓

Each term is tripled: geometric, $r = 3$.

80 State whether 90, 80, 70, 60, ... is arithmetic or geometric, and find its 12th term.

a. Arithmetic, -20 ✓

b. Arithmetic, -10

c. Arithmetic, 30

d. Geometric, -20

$d = -10$: $T_{12} = 90 + 11(-10) = -20$.

81 State whether 5, 10, 20, 40, ... is arithmetic or geometric, and give the common difference or ratio.

a. Geometric, $r = 2$ ✓

b. Geometric, $r = 5$

c. Arithmetic, $d = 2$

d. Arithmetic, $d = 5$

Each term is doubled: geometric, $r = 2$.

MCQ — Reading a sequence from its terms

82 The bar chart shows the first five terms of a sequence. Find the common difference.

a. 6

b. 5 ✓

c. 4

d. 9

Each term increases by 5.

83 Using the same bar chart, predict the 8th term.

a. 29

b. 34

c. 39 ✓

d. 44

$d = 5, a = 4: T_8 = 4 + 7(5) = 39.$

84 Using the same bar chart, find the sum of the first five terms shown.

a. 70 ✓

b. 66

c. 24

d. 75

$4 + 9 + 14 + 19 + 24 = 70.$

MCQ — Word problems: sequences in context

- 85 An employee's starting monthly salary is 3,500 dollars, and it increases by 150 dollars each year. Find the salary in year 6.
- a. 3,950 dollars
 - b. 4,400 dollars
 - c. 4,100 dollars
 - d. 4,250 dollars ✓

$$a = 3,500, d = 150: T_6 = 3,500 + 5(150) = 4,250.$$

- 86 A bacteria culture starts with 300 cells and doubles every hour. Find the number of cells after 5 hours.
- a. 19,200
 - b. 1,500
 - c. 9,600 ✓
 - d. 4,800

$$300 \times 2^5 = 300 \times 32 = 9,600.$$

- 87 A theater's ticket sales grow according to a geometric sequence: 200 tickets sold on day 1, with a common ratio of 1.5. Find the number of tickets sold on day 4.
- a. 900
 - b. 675 ✓
 - c. 300
 - d. 450

$$T_4 = 200(1.5)^3 = 200(3.375) = 675.$$

Multiple choice — easy

88 Find the 9th term of the arithmetic sequence with $a = -10$, $d = 2$.

a. 4

b. 9

c. 8

d. 6 ✓

$$a_9 = -10 + 8(2) = -10 + 16 = 6.$$

89 Find the 16th term of the arithmetic sequence with $a = 20$, $d = -7$.

a. -82

b. -92

c. -85 ✓

d. -78

$$a_{16} = 20 + 15((-7)) = 20 - 105 = -85.$$

90 Find the 14th term of the arithmetic sequence with $a = 6$, $d = 2$.

a. 35

b. 32 ✓

c. 34

d. 30

$$a_{14} = 6 + 13(2) = 6 + 26 = 32.$$

91 Find the 5th term of the arithmetic sequence with $a = -19$, $d = -5$.

a. -36

b. -39 ✓

c. -44

d. -34

$$a_5 = -19 + 4((-5)) = -19 - 20 = -39.$$

92 Find the 16th term of the arithmetic sequence with $a = -14$, $d = 5$.

a. 56

b. 61 ✓

c. 66

d. 64

$$a_{16} = -14 + 15(5) = -14 + 75 = 61.$$

93 Find the 13th term of the arithmetic sequence with $a = -16$, $d = 9$.

a. 95

b. 101

c. 83

d. 92 ✓

$$a_{13} = -16 + 12(9) = -16 + 108 = 92.$$

94 Find the 20th term of the arithmetic sequence with $a = -7$, $d = -7$.

a. -133

b. -140 ✓

c. -147

d. -137

$$a_{20} = -7 + 19((-7)) = -7 - 133 = -140.$$

95 Find the 10th term of the arithmetic sequence with $a = 12$, $d = 5$.

- a. 60
- b. 62
- c. 52
- d. 57 ✓

$$a_{10} = 12 + 9(5) = 12 + 45 = 57.$$

96 Find the 7th term of the arithmetic sequence with $a = 11$, $d = -5$.

- a. -14
- b. -19 ✓
- c. -24
- d. -16

$$a_7 = 11 + 6((-5)) = 11 - 30 = -19.$$

97 Find the 8th term of the arithmetic sequence with $a = 6$, $d = 2$.

- a. 23
- b. 22
- c. 18
- d. 20 ✓

$$a_8 = 6 + 7(2) = 6 + 14 = 20.$$

98 Find the 15th term of the arithmetic sequence with $a = -20$, $d = -8$.

- a. -129
- b. -132 ✓
- c. -124
- d. -140

$$a_{15} = -20 + 14((-8)) = -20 - 112 = -132.$$

99 Find the 10th term of the arithmetic sequence with $a = 12$, $d = -6$.

a. -42 ✓

b. -48

c. -39

d. -36

$$a_{10} = 12 + 9((-6)) = 12 - 54 = -42.$$

100 Find the 11th term of the arithmetic sequence with $a = -8$, $d = 1$.

a. 1

b. 3

c. 2 ✓

d. 5

$$a_{11} = -8 + 10(1) = -8 + 10 = 2.$$

101 Find the 17th term of the arithmetic sequence with $a = -4$, $d = 5$.

a. 71

b. 81

c. 76 ✓

d. 79

$$a_{17} = -4 + 16(5) = -4 + 80 = 76.$$

102 Find the 5th term of the arithmetic sequence with $a = -2$, $d = 9$.

- a. 25
- b. 43
- c. 37
- d. 34 ✓

$$a_5 = -2 + 4(9) = -2 + 36 = 34.$$

Multiple choice — medium

103 Find the 7th term of the geometric sequence with $a = -2$, $r = 2$.

- a. -131
- b. -128 ✓
- c. 128
- d. -125

$$a_7 = -2(2)^6 = -2(64) = -128.$$

104 Find the 3rd term of the geometric sequence with $a = -2$, $r = 2$.

- a. -8 ✓
- b. 8
- c. -11
- d. -5

$$a_3 = -2(2)^2 = -2(4) = -8.$$

105 Find the 4th term of the geometric sequence with $a = 3$, $r = -3$.

a. -81 ✓

b. 81

c. -78

d. -84

$$a_4 = 3((-3))^3 = 3((-27)) = -81.$$

106 Find the 6th term of the geometric sequence with $a = -9$, $r = -2$.

a. 291

b. -288

c. 285

d. 288 ✓

$$a_6 = -9((-2))^5 = -9((-32)) = 288.$$

107 Find the 4th term of the geometric sequence with $a = -7$, $r = -2$.

a. 56 ✓

b. 59

c. -56

d. 53

$$a_4 = -7((-2))^3 = -7((-8)) = 56.$$

108 Find the 6th term of the geometric sequence with $a = -6$, $r = 2$.

a. -195

b. 192

c. -192 ✓

d. -189

$$a_6 = -6(2)^5 = -6(32) = -192.$$

109 Find the 5th term of the geometric sequence with $a = 1$, $r = 3$.

a. 78

b. 81 ✓

c. -81

d. 84

$$a_5 = 1(3)^4 = 1(81) = 81.$$

110 Find the 4th term of the geometric sequence with $a = 1$, $r = -3$.

a. -27 ✓

b. -30

c. 27

d. -24

$$a_4 = 1((-3))^3 = 1((-27)) = -27.$$

111 Find the 3rd term of the geometric sequence with $a = -4$, $r = 2$.

a. -19

b. -13

c. 16

d. -16 ✓

$$a_3 = -4(2)^2 = -4(4) = -16.$$

112 Find the 5th term of the geometric sequence with $a = -2$, $r = 3$.

a. -162 ✓

b. -159

c. 162

d. -165

$$a_5 = -2(3)^4 = -2(81) = -162.$$

113 Find the 7th term of the geometric sequence with $a = -2$, $r = -3$.

a. 1458

b. -1461

c. -1455

d. -1458 ✓

$$a_7 = -2((-3))^6 = -2(729) = -1458.$$

114 Find the 3rd term of the geometric sequence with $a = -5$, $r = -3$.

a. -48

b. -42

c. -45 ✓

d. 45

$$a_3 = -5((-3))^2 = -5(9) = -45.$$

115 Find the 7th term of the geometric sequence with $a = -7$, $r = -2$.

a. 448

b. -445

c. -448 ✓

d. -451

$$a_7 = -7((-2))^6 = -7(64) = -448.$$

116 Find the 6th term of the geometric sequence with $a = -8$, $r = 2$.

- a. 256
- b. -253
- c. -259
- d. -256 ✓

$$a_6 = -8(2)^5 = -8(32) = -256.$$

117 Find the 3rd term of the geometric sequence with $a = -2$, $r = 3$.

- a. -15
- b. 18
- c. -21
- d. -18 ✓

$$a_3 = -2(3)^2 = -2(9) = -18.$$

Multiple choice — hard

118 Find the sum of the infinite geometric series with $a = 4$, $r = \frac{1}{2}$.

- a. 8 ✓
- b. 6
- c. -8
- d. 10

$$1 - r = 1 - \frac{1}{2} = \frac{1}{2}. S = \frac{a}{1 - r} = 4 \times \frac{2}{1} = \frac{8}{1} = 8.$$

119 Find the sum of the infinite geometric series with $a = 18$, $r = \frac{1}{3}$.

- a. 25
- b. 29
- c. 27 ✓
- d. -27

$$1 - r = 1 - \frac{1}{3} = \frac{2}{3}. S = \frac{a}{1 - r} = 18 \times \frac{3}{2} = \frac{54}{2} = 27.$$

120 Find the sum of the infinite geometric series with $a = 6$, $r = \frac{1}{3}$.

- a. 9 ✓
- b. 7
- c. 11
- d. -9

$$1 - r = 1 - \frac{1}{3} = \frac{2}{3}. S = \frac{a}{1 - r} = 6 \times \frac{3}{2} = \frac{18}{2} = 9.$$

121 Find the sum of the infinite geometric series with $a = 16$, $r = \frac{1}{4}$.

- a. $\frac{64}{3}$ ✓
- b. $-\frac{64}{3}$
- c. $\frac{58}{3}$
- d. $\frac{70}{3}$

$$1 - r = 1 - \frac{1}{4} = \frac{3}{4}. S = \frac{a}{1 - r} = 16 \times \frac{4}{3} = \frac{64}{3}.$$

122 Find the sum of the infinite geometric series with $a = 30$, $r = \frac{1}{5}$.

- a. $\frac{79}{2}$
- b. $\frac{75}{2}$ ✓
- c. $-\frac{75}{2}$
- d. $\frac{71}{2}$

$$1 - r = 1 - \frac{1}{5} = \frac{4}{5}. S = \frac{a}{1 - r} = 30 \times \frac{5}{4} = \frac{150}{4} = \frac{75}{2}.$$

123 Evaluate $\sum_{k=1}^5 (k + 4)$.

- a. 31
- b. -35
- c. 39
- d. 35 ✓

Terms: 5, 6, 7, 8, 9. Sum = 35.

124 Evaluate $\sum_{k=1}^6 (k - 2)$.

- a. -9
- b. 13
- c. 9 ✓
- d. 5

Terms: -1, 0, 1, 2, 3, 4. Sum = 9.

125 Evaluate $\sum_{k=1}^6 (2k + 2)$.

- a. -54
- b. 54 ✓
- c. 50
- d. 58

Terms: 4, 6, 8, 10, 12, 14. Sum = 54.

126 Evaluate $\sum_{k=1}^5 (2k + 2)$.

a. -40

b. 36

c. 40 ✓

d. 44

Terms: 4, 6, 8, 10, 12. Sum = 40.

127 Evaluate $\sum_{k=1}^6 (k + 5)$.

a. 55

b. 51 ✓

c. -51

d. 47

Terms: 6, 7, 8, 9, 10, 11. Sum = 51.

128 A sequence satisfies $a_1 = 6$ and $a_{n+1} = a_n + 2$. Find a_6 .

a. 14

b. 13

c. 16 ✓

d. 19

$a_1 = 6, a_2 = 8, a_3 = 10, a_4 = 12, a_5 = 14, a_6 = 16$.

129 A sequence satisfies $a_1 = 6$ and $a_{n+1} = 3a_n + 3$. Find a_4 .

a. 201 ✓

b. 198

c. 66

d. 204

$a_1 = 6, a_2 = 21, a_3 = 66, a_4 = 201$.

130 A sequence satisfies $a_1 = 2$ and $a_{n+1} = a_n + 4$. Find a_4 .

a. 10

b. 14 ✓

c. 17

d. 11

$$a_1 = 2, a_2 = 6, a_3 = 10, a_4 = 14.$$

131 A sequence satisfies $a_1 = 4$ and $a_{n+1} = 3a_n - 1$. Find a_6 .

a. 854

b. 284

c. 851 ✓

d. 848

$$a_1 = 4, a_2 = 11, a_3 = 32, a_4 = 95, a_5 = 284, a_6 = 851.$$

132 A sequence satisfies $a_1 = 7$ and $a_{n+1} = a_n + 3$. Find a_6 .

a. 22 ✓

b. 27

c. 25

d. 19

$$a_1 = 7, a_2 = 10, a_3 = 13, a_4 = 16, a_5 = 19, a_6 = 22.$$